25901051 **Reg.No: ……**

Name: ……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCREM100 - ELECTRICAL FUNDAMENTALS AND ELECTRONICS

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **25** questions

Each question carries **1** mark.

1. Show the dielectric material in capacitors. [K]/[CO1]

Ans: In a capacitor, the dielectric material is the insulating medium placed between the two plates such as air, mica, paper, plastic, ceramic, or glass, which increases the capacitance.

2. Recall the unit of resistance. [K]/[CO1]

Ans: The unit of resistance is ohm (Ω).

3. Define a relay. [K]/[CO1]

Ans: A relay is an electromagnetic switch used to control a high-power circuit using a low-power signal.

4. Recognize the effect of increasing inductance on current. [K]/[C01]

Ans: When inductance increases, the current decreases or opposes the change in current.

5. Identify another name for variable resistor. [K]/[C01]

Ans: Another name for a variable resistor is potentiometer.

6. Identify the construction parts of a coaxial cable. [K]/[C01]

Ans: A coaxial cable has two main parts: inner conductor and outer shield (separated by an insulating layer).

7. List two examples of voltage sources. [K]/[CO1]

Ans: Battery and Generator.

8. Identify the unit of inductance. [U]/[CO2]

Ans: The unit of inductance is Henry (H).

9. Classify the components of an LCR circuit. [A]/[CO3]

Ans: Inductor (L), Capacitor (C), and Resistor (R).

10. Explain susceptance. [U]/[CO2]

Ans: Susceptance is the reciprocal of reactance and it indicates how easily a circuit allows AC to pass.

11. Write the full form of LED. [A]/[CO4]

Ans: Light Emitting Diode.

12. Write the formula for power in AC circuits [A]/[CO3]

Ans: $P = VI \cos \phi$

13. Differentiate between AC and DC. [U]/[CO2]

Ans: AC changes direction periodically, while DC flows in one fixed direction.

14. Write the equation of a sinusoidal AC voltage [A]/[CO3]

Ans: v = V□ sin(ωt)

15. Discuss the main purpose of Norton's theorem. [U]/[CO2]

Ans: The main purpose of Norton's theorem is to simplify a complex network into an equivalent current source in parallel with a resistance.

16. Write the power in a three-phase system. [A]/[CO3]

Ans: P = √3 V□ I□ cos ϕ

17. Present one example of a semiconductor material. [A]/[CO5]

Ans: One example of a semiconductor material is Silicon (Si)

18. Indicate one electrical property of metals. [U]/[CO4]

Ans: Metals have high electrical conductivity.

19. Write the three terminals of a transistor. [A]/[CO5]

Ans: Emitter (E), Base (B), Collector (C).

20. Illustrate forward biasing in a P-N junction. [A]/[CO4]

Ans: In forward biasing, the P-region of a PN junction is connected to the positive terminal of the battery and the N-region to the negative terminal, allowing current to flow through the junction.

21. Show what an amplifier increases in a signal. [A]/[CO5]

Ans: An amplifier increases the amplitude (strength) of a signal.

22. Explain the type of current allowed by a diode [U]/[CO4]

Ans: A diode allows current to flow in only one direction (forward direction) and blocks it in the reverse direction.

23. Write one example of an electrical insulator. [A]/[CO5]

Ans: One example of an electrical insulator is Rubber.

24. Describe conduction electrons. [U]/[CO4]

Ans: Conduction electrons are the free electrons in a material that carry electric current.

25. Write the concept of a half-wave rectifier. [A]/[CO5]

Ans: A half-wave rectifier allows only one half (positive or negative) of an AC input to pass, converting AC into pulsating DC.

26. Explain the concept of doping in semiconductors. [U]/[CO4]

Ans: Doping is the process of adding impurities to a pure semiconductor to increase its conductivity.

27. Write the meaning of mass number. [A]/[CO5]

Ans: Mass number is the total number of protons and neutrons in the nucleus of an atom.

[1X25=25]

Part B

Short Essay

Answer any **5** questions

Each question carries **5** marks

28. State the construction of relay. [K]/[CO1]

Ans: A relay is an important electromechanical device used in electrical and electronic circuits for controlling high power or high voltage circuits with the help of a small control signal. The construction of a relay is quite simple but very effective. At the heart of a relay lies an electromagnet, which is basically a coil of insulated copper wire wound around a soft iron core. Along with this, there is an armature, which is a movable soft iron piece connected to a spring. The armature is placed close to the electromagnet in such a way that when the coil is energized, the magnetic field pulls the armature towards the core. The armature is mechanically linked with a set of contacts. These contacts are of two types—normally open (NO) and normally closed (NC). A base frame holds all these parts together, and the spring helps the armature return to its original position when the coil is de-energized**.**

29. List the different types of switches and explain toggle switch. [K]/[CO1]

Ans: Types of switches ; Slide Switch, toggle switch, push to, on switch and push to off switch.

A toggle switch is one of the most common types of electrical switches used to control the flow of current in a circuit. It operates through a simple mechanism where a lever, handle, or knob is flipped back and forth to make or break an electrical connection. The name “toggle” comes from the mechanical action of moving the lever between two stable positions, usually ON and OFF. This simple yet reliable design has made toggle switches highly popular in a wide variety of electrical and electronic applications. Structurally, a toggle switch consists of a lever connected to an internal set of electrical contacts. When the lever is pushed in one direction, the contacts close and allow current to pass through, whereas when the lever is pushed in the opposite direction, the contacts open and stop the flow of electricity. Depending on the requirement, toggle switches can be single-pole single-throw (SPST), single-pole double- throw (SPDT), double-pole single-throw (DPST), or double-pole double-throw (DPDT). This variety allows them to be used for simple ON/OFF control as well as for more complex switching operations like reversing the direction of a motor or selecting between two different circuits. One of the key advantages of toggle switches is their durability and ease of use. The firm clicking action when the lever is flipped provides tactile feedback, making them convenient to operate even without looking. They are also capable of handling a wide range of voltages and currents, which makes them suitable for both low-power electronic devices and heavy-duty electrical equipment. Moreover, toggle switches are available in different sizes, mounting styles, and designs, making them adaptable to various applications in industries, homes, and vehicles. From a practical standpoint, toggle switches are essential in daily life because they provide a simple, mechanical, and reliable means of controlling electricity. Their visible ON/OFF positions also make them safe and easy to identify, reducing the risk of accidental misuse.

30. Explain the application of LCR circuits in resonance experiments and power factor improvement.

[U]/[CO2]

Ans: LCR circuits (containing an inductor L, capacitor C, and resistor R) are widely used in resonance experiments because they show a sharp rise in current at the resonant frequency. At resonance, the inductive reactance and capacitive reactance cancel each other, making the circuit purely resistive. This principle is used in radio receivers, communication systems, and frequency-selective filters to select a particular frequency from many incoming signals.

LCR circuits are also useful in improving the power factor of electrical systems. Industrial loads often have low power factor due to inductive machines. By connecting an LCR circuit—usually a capacitor in series or parallel with the load—the reactive power is reduced and the circuit becomes closer to resistive. This improves the power factor, reduces losses, and increases the efficiency of power transmission.

31. Explain the working of a Zener diode as a voltage regulator. [U]/[CO4]

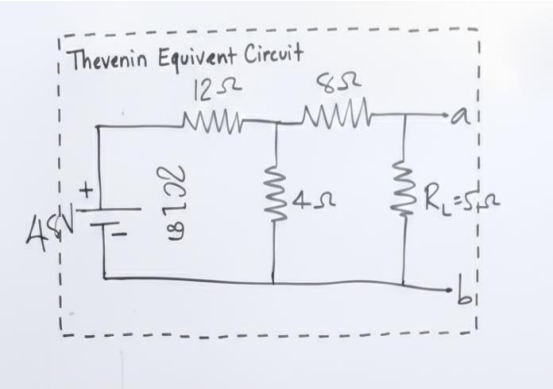
Ans: A Zener diode is a special type of diode that is mainly designed to work in the reverse bias condition. In an ordinary diode, when reverse voltage increases beyond a certain limit, the diode breaks down permanently and gets damaged. But in a Zener diode, the diode is specially made so that it can safely undergo this breakdown without getting destroyed. This safe breakdown is known as Zener breakdown.
The most important feature of a Zener diode is that it maintains a constant voltage across it,

even if the current flowing through it changes. Because of this property, it is widely used as a voltage regulator in electronic circuits. The voltage at which the Zener diode breaks down and starts conducting in reverse bias is called the Zener voltage. This voltage is a fixed value, and it is carefully set during the manufacturing process. For example, a 5.6V Zener diode will always maintain about 5.6V across it once it enters breakdown. To understand its working, imagine a circuit where the Zener diode is connected in reverse across a power supply. As the supply voltage increases, initially the diode does not conduct much current. But once the supply reaches the Zener voltage, the diode suddenly allows current to flow in reverse. At the same time, the voltage across the diode does not increase further; it stays nearly constant at the Zener voltage. In this way, the Zener diode protects sensitive components in the circuit from high voltages and ensures that they always receive a stable supply.

The operation of the Zener diode is based on two mechanisms: Zener breakdown and avalanche breakdown. Zener breakdown happens at low reverse voltages, where the electric field is strong enough to pull electrons from their atomic bonds. Avalanche breakdown happens at higher voltages, where carriers gain enough energy to knock out more electrons and create a chain reaction. In both cases, the diode can conduct in reverse safely, but without being damaged.
Zener diodes are manufactured in different voltage ratings, from a few volts up to several hundred volts. They are usually made from silicon because silicon provides better performance and stability. When used in circuits, a small resistor is generally connected in series with the Zener diode. This resistor limits the current flowing through the diode and prevents it from overheating. In practice, Zener diodes are commonly used for voltage regulation in power supplies, protection of circuits from over-voltage, waveform shaping, and reference voltage generation. For example, in a power supply circuit, if we use a 12V Zener diode, then no matter how much the input supply fluctuates above 12V, the output will remain fixed at 12V. This makes Zener diodes very important in designing stable electronic systems. Thus, the Zener diode is not just a normal diode but a specially designed component that takes advantage of controlled reverse breakdown. Its ability to maintain constant voltage makes it one of the simplest and most effective devices for regulation and protection in electronics.

32. Demonstrate how to find the Thevenin equivalent of a simple network. [U]/[CO2]

Ans:



With RL removed there is no current through the 8 Ω, so node a= node N(the junction after 12 Ω). That node is a simple divider between 12 Ω and 4 Ω:

$$V_{th} = 45 \cdot \frac{4}{12+4} = 45 \cdot \frac{4}{16} = 45 \cdot 0.25 = 11.25 \text{ V}.$$

Thevenin resistance (Rth):
Turn the 45 V source off (short it). Then the 12 Ω and 4 Ω appear in parallel from node $N$to ground:

$$R_N = \frac{12 \cdot 4}{12+4} = 3\ \Omega,$$

and this is in series with the 8 Ω between $a$and N, so

$$R_{th} = 8 + 3 = 11\ \Omega.$$

Answer: $V_{th} = 11.25$ V, $R_{th} = 11\ \Omega$.

(If you reconnect $R_L = 5\ \Omega$: $I_L = \frac{11.25}{11+5} = 0.703$ A, $V_L = I_L \cdot 5 \approx 3.52$ V.)

33. Describe the structure and behavior of electrons in an atom. [U]/[CO4]

Ans: An electron is one of the fundamental particles that make up all matter in the universe. It is a negatively charged subatomic particle found in every atom. Electrons play a vital role in almost all physical and chemical processes, from electricity and magnetism to chemical bonding and atomic structure. The discovery of the electron was a major milestone in science, as it helped scientists understand the nature of matter and the structure of the atom.
Electrons are extremely small in both mass and size. The mass of an electron is approximately 9.11 × 10−31 kilograms, and its charge is –1.6 × 10−19 coulombs. The charge of an electron is equal in magnitude but opposite in sign to the charge of a proton. Because electrons carry a negative charge, they are attracted to positively charged particles like protons and repelled by other electrons.
In an atom, electrons are found in specific energy levels or shells surrounding the nucleus. The nucleus, which contains protons and neutrons, is at the center of the atom, and electrons revolve around it in different orbits. These energy levels are arranged in such a way that the electrons closer to the nucleus have less energy, and those farther away have more energy. The arrangement of electrons in these shells determines the chemical properties of an element.
When atoms interact or bond with one another, it is mainly the electrons in the outermost shell, known as valence electrons, that are involved.Electrons are also responsible for electric current. In metals and conductors, electrons are free to move from one atom to another. When a potential difference (voltage) is applied across a conductor, these free electrons drift in one direction, forming an electric current. This flow of electrons is the basis of all electrical and electronic devices, from simple bulbs to complex computers. Electrons can also behave like both particles and waves, a property known as wave-particle duality. This means that in some experiments, electrons act as tiny particles, while in others,they show wave-like behavior. This dual nature was confirmed by the Davisson-Germer experiment, which showed that electrons could produce diffraction patterns like light waves.
In modern technology, electrons have countless applications. They are used in television and

computer screens, microscopes, vacuum tubes, semiconductors, and electronic circuits. The movement and control of electrons are the foundation of the entire field of electronics. Even in communication systems and computing, it is the precise manipulation of electrons that allows information to be processed and transmitted.
The electron is one of the most essential building blocks of matter and technology. It plays a fundamental role in atomic structure, chemical reactions, electricity, and modern electronic devices. The discovery and understanding of electrons not only transformed physics and chemistry but also laid the foundation for the technological world we live in today.

34. Classify the different types of phototransistors and present their working differences based on sensitivity and response time. [A]/[CO5]

Ans: Phototransistors are classified mainly into Bipolar Junction Phototransistors (BJT phototransistors) and Field Effect Phototransistors (FET-based or MOSFET phototransistors). They may further be categorized as phototransistors with base terminal, phototransistors without base terminal, and Darlington phototransistors.

1. BJT Phototransistors:
   These are the most common type and work on the principle that incident light generates photocurrent at the base–collector junction. They offer high sensitivity because light acts as the base current that is amplified by transistor gain. However, due to this large internal capacitance, they have a slower response time, making them unsuitable for very high-speed applications.
2. MOSFET / FET-based Phototransistors:
   These devices use light to modulate the channel conductivity. They have very fast response times because of low junction capacitance, making them suitable for high-speed optical communication. Their sensitivity is lower compared to BJT phototransistors, as they do not provide current gain through transistor action.
3. Darlington Phototransistors:
   Formed by connecting two phototransistors in Darlington configuration, they provide extremely high sensitivity due to double current amplification. However, they exhibit the slowest response time among all types because of increased capacitance and charge-storage effects.
4. Base-terminal vs. Base-less Phototransistors:
   - With base terminal: Sensitivity can be externally controlled by applying a bias at the base, but response time is slightly slower.
   - Without base terminal: Fixed sensitivity but faster response due to reduced capacitance.

[5x5 = 25]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

35. Define a rocker switch and state its construction and working principle with a neat diagram.

[K]/[C01]

Ans: A rocker switch is an electromechanical switching device that operates by rocking a movable actuator back and forth to open or close an electrical circuit. It is widely used in household appliances, power supplies, automotive dashboards and control panels because it provides a simple ON/OFF action with clear visual and tactile feedback. When one side of the actuator is pressed, the switch rocks into the ON position and completes the circuit; when the opposite side is pressed, it rocks back to the OFF position and breaks the circuit. This rocking motion gives the switch its name and makes it easy to operate even without looking at it.

In terms of construction, a rocker switch consists of a moulded insulating housing that encloses all the electrical and mechanical components. The upper part of the device is the rocker or actuator, which is shaped like a small curved plate that pivots around a central fulcrum. Inside the switch, a small spring mechanism is provided to give a firm snap action and to hold the switch reliably in either the ON or OFF position. Beneath the actuator lie the metallic contacts. These include one or more fixed contacts connected to the external terminals and a moving contact that is attached to the rocker mechanism. When the actuator is rocked, the moving contact either touches the fixed contacts to complete the circuit, or lifts away to open the circuit. The terminals at the bottom of the switch allow it to be connected to external wiring, typically in SPST or SPDT configurations. The entire assembly is compact, insulated and designed to withstand repeated mechanical operation.

The working principle of a rocker switch is based purely on mechanical contact-making and breaking. When the user presses the rocker on the ON side, the actuator pivots around its hinge and pushes the moving contact into the fixed contacts. This closes the circuit and allows the current to flow. The spring provides stability so that the switch remains in this position until it is manually pressed again. Pressing the opposite side of the rocker causes the actuator to reverse its position, and the moving contact is pulled away from the fixed contacts, thereby interrupting the current flow and switching the device OFF. Some rocker switches include an internal indicator lamp or LED that glows when the circuit is closed, making it easy to identify the switch state.

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36. Write short notes on practical applications of series and parallel LCR circuits, including Q-factor and power factor. [A]/[CO3]

Ans: Series and parallel LCR circuits play an important role in many practical electronic and electrical applications because of their ability to control impedance, resonance, selectivity, and phase relationships between voltage and current. A series LCR circuit combines an inductor, capacitor and resistor in a single path. When such a circuit is driven by an AC source, it exhibits a sharp resonance at the frequency where the inductive reactance equals the capacitive reactance. At this resonant condition, the impedance of the circuit becomes minimum and the current reaches a maximum value. This property makes series LCR circuits extremely useful in applications such as radio tuning circuits, signal selection in communication receivers, induction heating, and audio frequency filtering. The circuit's Q-factor, defined as the ratio of reactive power to real power at resonance, determines the sharpness of resonance. A high Q-factor indicates high selectivity, meaning that the circuit can isolate a narrow band of frequencies, which is essential in radio broadcast receivers and selective amplifiers.

A parallel LCR circuit, on the other hand, consists of the resistor, inductor and capacitor connected in parallel across an AC supply. Instead of minimizing impedance at resonance

like the series circuit, the parallel arrangement produces a condition where the total current drawn from the source becomes minimum at the resonant frequency. The circuit impedance is maximum at resonance, and most of the current circulates between the inductor and capacitor branches. This makes parallel LCR circuits suitable for applications like tank circuits in oscillators, impedance matching networks, and filter networks where stable frequency operation and high impedance are required. The Q-factor of a parallel LCR circuit also influences the stability and purity of oscillations produced in LC oscillators and RF transmitters, as a higher Q ensures a narrower bandwidth and better frequency stability.

In both types of circuits, the power factor plays a significant role in determining energy efficiency. Power factor is the cosine of the phase angle between the supply voltage and the current. In LCR circuits, the presence of inductive and capacitive reactance causes the current to lead or lag the voltage depending on which reactance dominates. At resonance, the power factor becomes unity because the net reactance is zero and the voltage and current are in phase. This condition is desirable in many power applications since it ensures maximum power transfer and minimum losses. Improving power factor in practical systems, such as transmission lines and industrial loads, often involves using LCR combinations to balance reactive power.

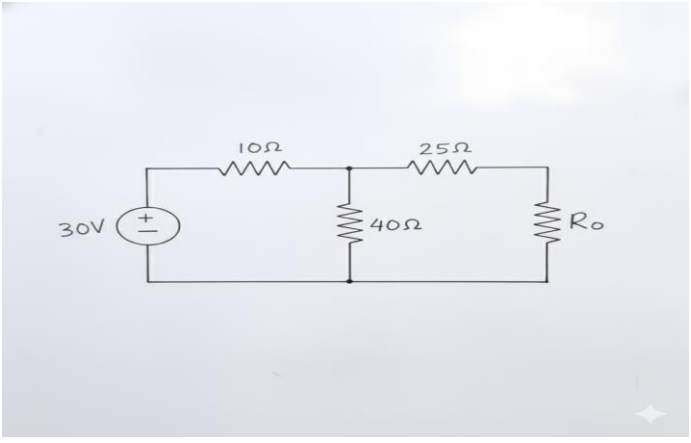
Thus, series and parallel LCR circuits find extensive use in tuning, filtering, signal processing, oscillation generation, and power management. Their behaviour, governed by resonance, Q-factor and power factor, makes them indispensable components in both communication electronics and power engineering applications.

37. Model a simple circuit and show how the maximum power transfer theorem is applied to find maximum power delivered to a load. [A]/[CO3]

Ans: The maximum power transfer theorem says that a load receives maximum power from a network when the load resistance (RL) is equal to the internal resistance (Rth) of the source.

In very simple words:

“A load gets maximum power when RL = Rth.”



Step 1 — Find V_{th}(open-circuit voltage at the load terminals):
Remove R_o(open circuit). Then no current flows in the 25Ωbranch, so the top junction left of the 25Ω(call it node A) is determined only by the $10\Omega$and the $40\Omega$to ground:

$$V_A = 30 \cdot \frac{40}{10+40} = 30 \cdot \frac{40}{50} = 30 \cdot 0.8 = 24\ \text{V}.$$

With no current through $25\Omega$the drop across it is zero, so the open terminal (node B) is at the same potential:

$$V_{th} = V_B = 24\ \text{V}.$$

Step 2 — Find R_{th}(with source turned off):
Short the 30 V source. Node A then has two resistances to ground: 10Ω(shorted source side) and 40Ω— these are in parallel:

$$R_{A\to g} = 10 \parallel 40 = \frac{10 \cdot 40}{10+40} = \frac{400}{50} = 8\ \Omega.$$

Looking into the open terminals (node B to ground) you see the 25Ωin series with that parallel combination, so

$$R_{th} = 25 + 8 = 33\ \Omega.$$

Step 3 — Maximum power delivered to the load:
Maximum power occurs when $R_L = R_{th} = 33\ \Omega$

$$P_{\max} = \frac{V_{th}^2}{4R_{th}} = \frac{24^2}{4 \cdot 33} = \frac{576}{132} \approx 4.36\ \text{W}.$$

- $V_{th} = 24$ V
- $R_{th} = 33\ \Omega$
- Optimum load $R_L = 33$ for maximum power
- $P_{\max} \approx 4.36$ W

38. Explain what valence electrons are and describe how they help in forming bonds between atoms.

[U]/[CO4]

Ans: Every atom in the universe is made up of three main particles — protons, neutrons, and electrons. Protons and neutrons stay inside the center of the atom called the nucleus, while electrons move around the nucleus in different circular paths known as shells or energy levels. These shells are named as K, L, M, N, and so on. Each shell can hold a certain number of electrons.Among all these electrons, the ones that are present in the outermost shell of an atom are called valence electrons. These electrons play a very important role in deciding how an atom will behave or react with other atoms. In simple words, valence electrons are responsible

for forming bonds between atoms.
For example, if we take a sodium atom, it has one electron in its outermost shell. That means it has one valence electron. Similarly, chlorine has seven electrons in its outermost shell. Sodium wants to lose one electron to become stable, and chlorine wants to gain one electron to complete its shell. When they come together, sodium gives its one valence electron to chlorine, and both become stable. This forms a compound called sodium chloride (NaCl) — which is nothing but common salt.
This shows that valence electrons decide whether an atom will give or take electrons and how it will combine with other atoms. If an atom has one, two, or three valence electrons, it usually tries to lose them. If it has five, six, or seven valence electrons, it usually tries to gain electrons. Atoms having eight valence electrons (like noble gases) are already stable and do not easily react with other elements.
Valence electrons also determine many other properties of an element, such as whether it is a metal, non-metal, or semiconductor. In metals, valence electrons are loosely bound to the nucleus and can move freely, which makes metals good conductors of electricity. In non-metals, valence electrons are tightly held, so they do not allow current to pass easily.
Valence electrons are the outermost electrons of an atom that decide its chemical behavior, bonding nature, and electrical properties. Without understanding valence electrons, we cannot explain why elements form compounds or how materials conduct electricity. They are the key to understanding the entire chemistry of atoms and the formation of all substances around us.

[10x2 = 20]